

TECHSHIP INTERNSHIP

Day 2 – JavaScript Basics

Submitted by : Janet Naveena

Date : 02-06-2026

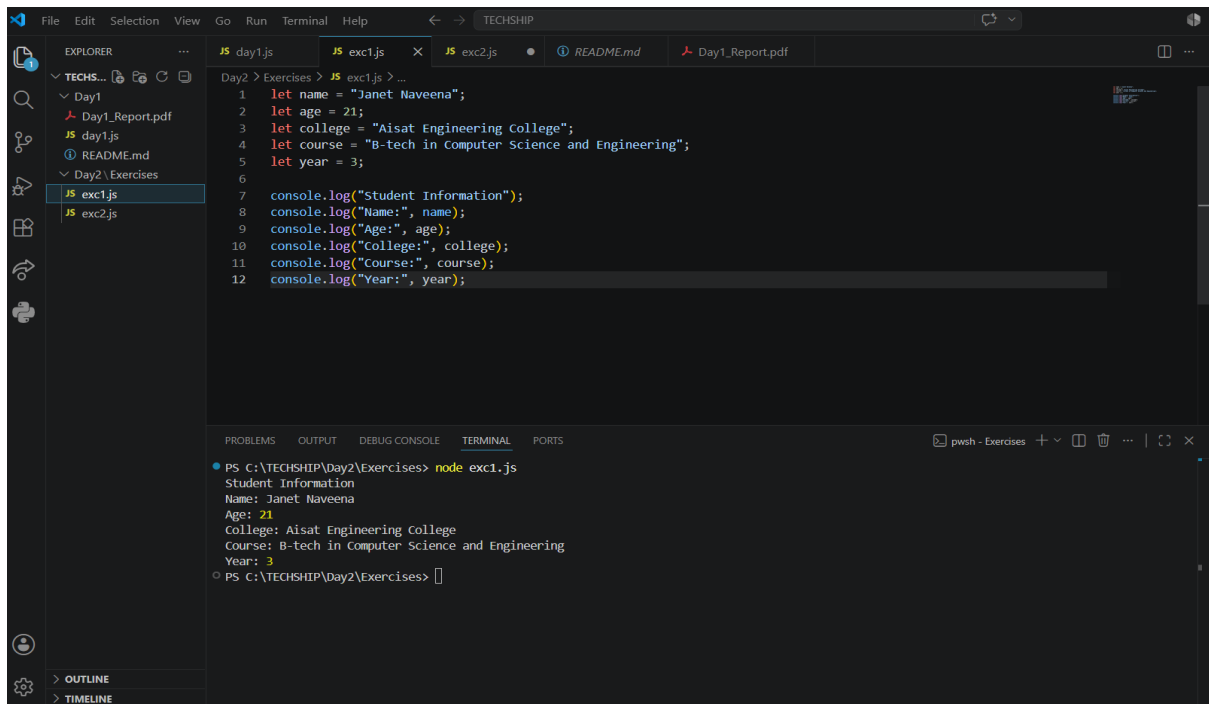
On Day 2 of the TECHSHIP Internship Program, I learned the fundamentals of JavaScript. The topics covered included variables, data types, operators, conditional statements, and loops. These concepts helped me understand how JavaScript is used to store data, perform calculations, make decisions, and automate repetitive tasks.

As part of the practical exercises and assignment, I developed several programs that involved calculations, decision-making, and repetitive operations. These activities provided hands-on experience and improved my understanding of programming logic and problem-solving techniques. I also learned the difference between numerical addition and string concatenation, which helped me better understand how JavaScript handles different data types.

One challenge I faced was understanding loop execution and debugging errors in the programs. Through practice and testing different examples, I was able to overcome these difficulties. The most interesting concepts were loops and conditional statements because they are useful for solving real-world problems. Overall, Day 2 strengthened my foundation in JavaScript and improved my confidence in writing basic programs.

Exercises

Exercise 1 – Student Information System



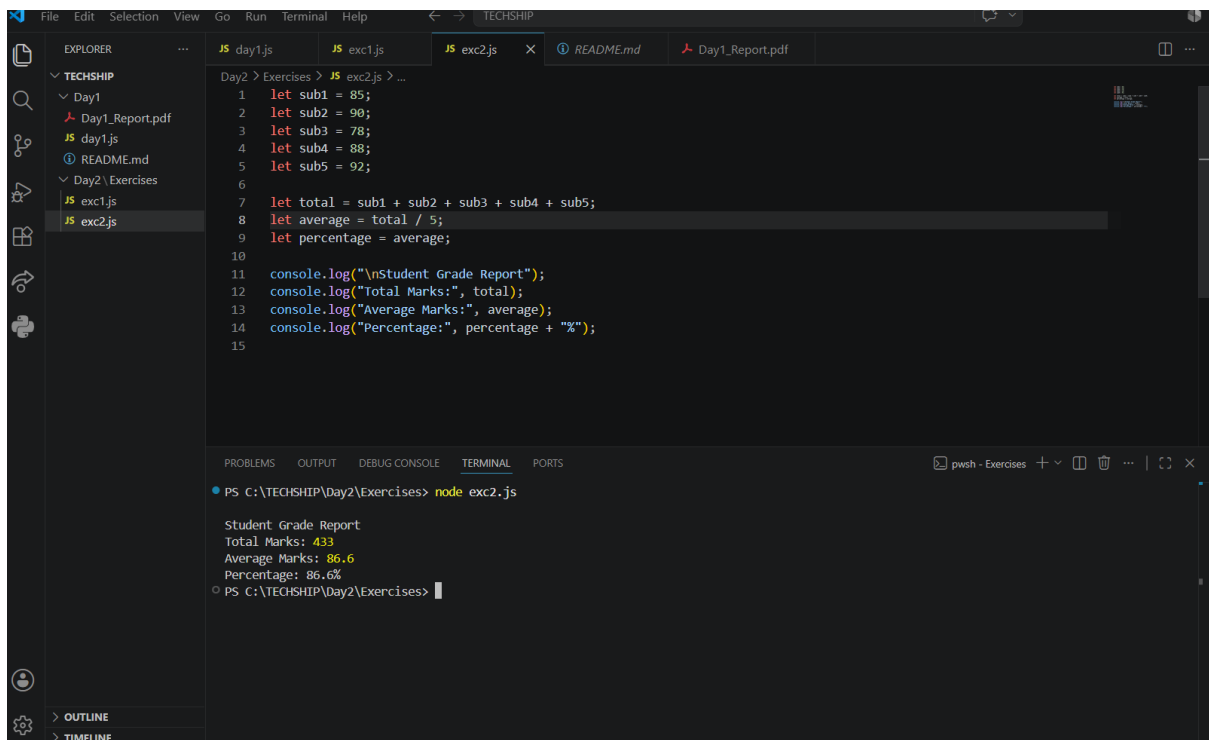
The screenshot shows the Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows the file structure: 'Day1' (containing 'Day1_Report.pdf') and 'Day2\Exercises' (containing 'exc1.js' and 'exc2.js'). The main editor window displays the code for 'exc1.js':

```
1 let name = "Janet Naveena";
2 let age = 21;
3 let college = "Aisat Engineering College";
4 let course = "B-tech in Computer Science and Engineering";
5 let year = 3;
6
7 console.log("Student Information");
8 console.log("Name:", name);
9 console.log("Age:", age);
10 console.log("College:", college);
11 console.log("Course:", course);
12 console.log("Year:", year);
```

Below the editor, the TERMINAL panel shows the command 'node exc1.js' being executed in a PowerShell window. The output is as follows:

```
PS C:\TECHSHIP\Day2\Exercises> node exc1.js
Student Information
Name: Janet Naveena
Age: 21
College: Aisat Engineering College
Course: B-tech in Computer Science and Engineering
Year: 3
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 2 – Student Grade Calculator



The screenshot shows the Visual Studio Code editor with the same 'TECHSHIP' project. The Explorer sidebar now highlights 'exc2.js' in the 'Day2\Exercises' folder. The main editor window displays the code for 'exc2.js':

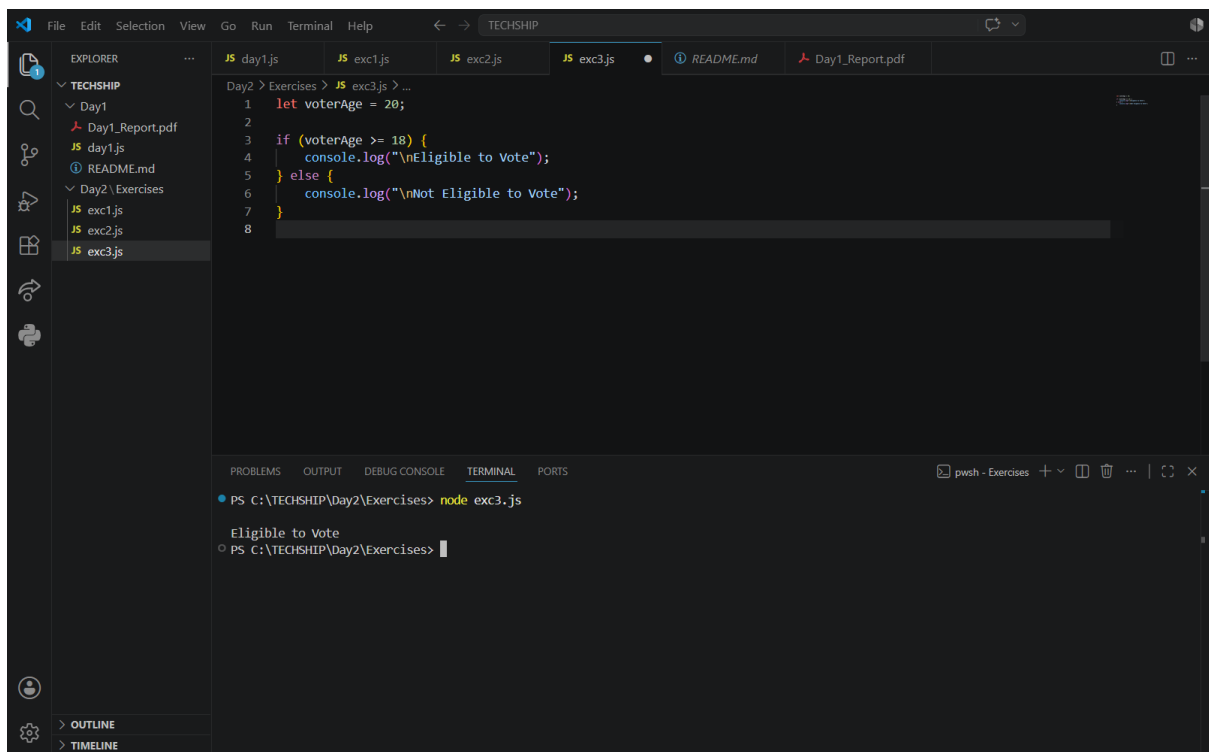
```
1 let sub1 = 85;
2 let sub2 = 90;
3 let sub3 = 78;
4 let sub4 = 88;
5 let sub5 = 92;
6
7 let total = sub1 + sub2 + sub3 + sub4 + sub5;
8 let average = total / 5;
9 let percentage = average;
10
11 console.log("\nStudent Grade Report");
12 console.log("Total Marks:", total);
13 console.log("Average Marks:", average);
14 console.log("Percentage:", percentage + "%");
15
```

The TERMINAL panel shows the command 'node exc2.js' being executed. The output is as follows:

```
PS C:\TECHSHIP\Day2\Exercises> node exc2.js

Student Grade Report
Total Marks: 433
Average Marks: 86.6
Percentage: 86.6%
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 3 – Voting Eligibility Checker



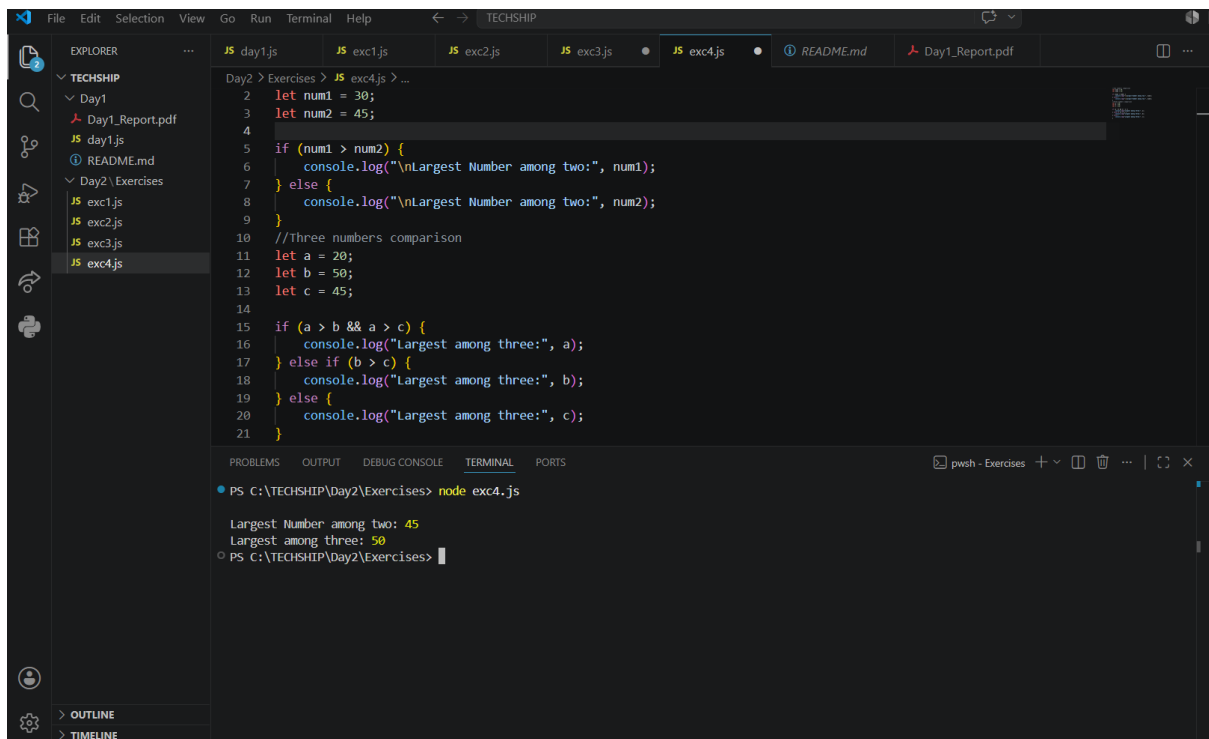
The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure. The main editor window shows the file `exc3.js` with the following JavaScript code:

```
1 let voterAge = 20;
2
3 if (voterAge >= 18) {
4   console.log("\nEligible to Vote");
5 } else {
6   console.log("\nNot Eligible to Vote");
7 }
8
```

The TERMINAL panel at the bottom shows the command prompt output after running `node exc3.js`:

```
PS C:\TECHSHIP\Day2\Exercises> node exc3.js
Eligible to Vote
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 4 – Largest Number Finder



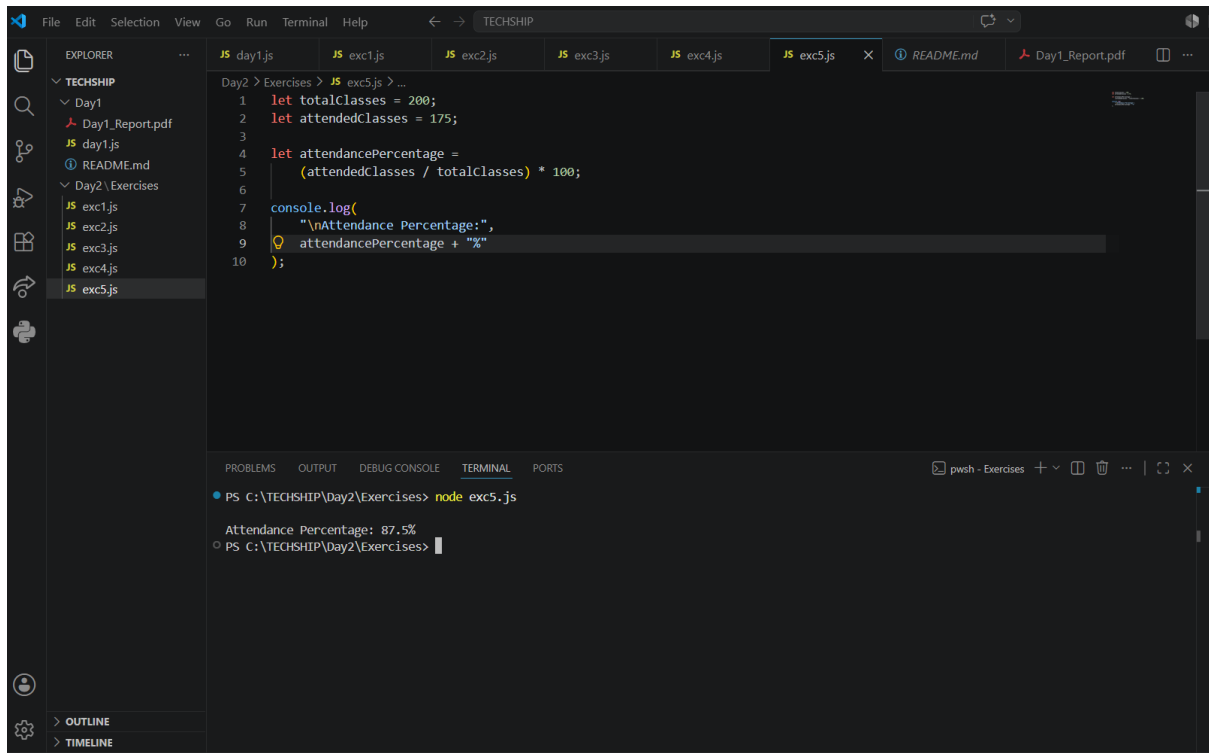
The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a project structure. The main editor window shows the file `exc4.js` with the following JavaScript code:

```
2 let num1 = 30;
3 let num2 = 45;
4
5 if (num1 > num2) {
6   console.log("\nLargest Number among two:", num1);
7 } else {
8   console.log("\nLargest Number among two:", num2);
9 }
10 //Three numbers comparison
11 let a = 20;
12 let b = 50;
13 let c = 45;
14
15 if (a > b && a > c) {
16   console.log("Largest among three:", a);
17 } else if (b > c) {
18   console.log("Largest among three:", b);
19 } else {
20   console.log("Largest among three:", c);
21 }
22
```

The TERMINAL panel at the bottom shows the command prompt output after running `node exc4.js`:

```
PS C:\TECHSHIP\Day2\Exercises> node exc4.js
Largest Number among two: 45
Largest among three: 50
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 5 – Attendance Calculator



```
File Edit Selection View Go Run Terminal Help
TECHSHIP

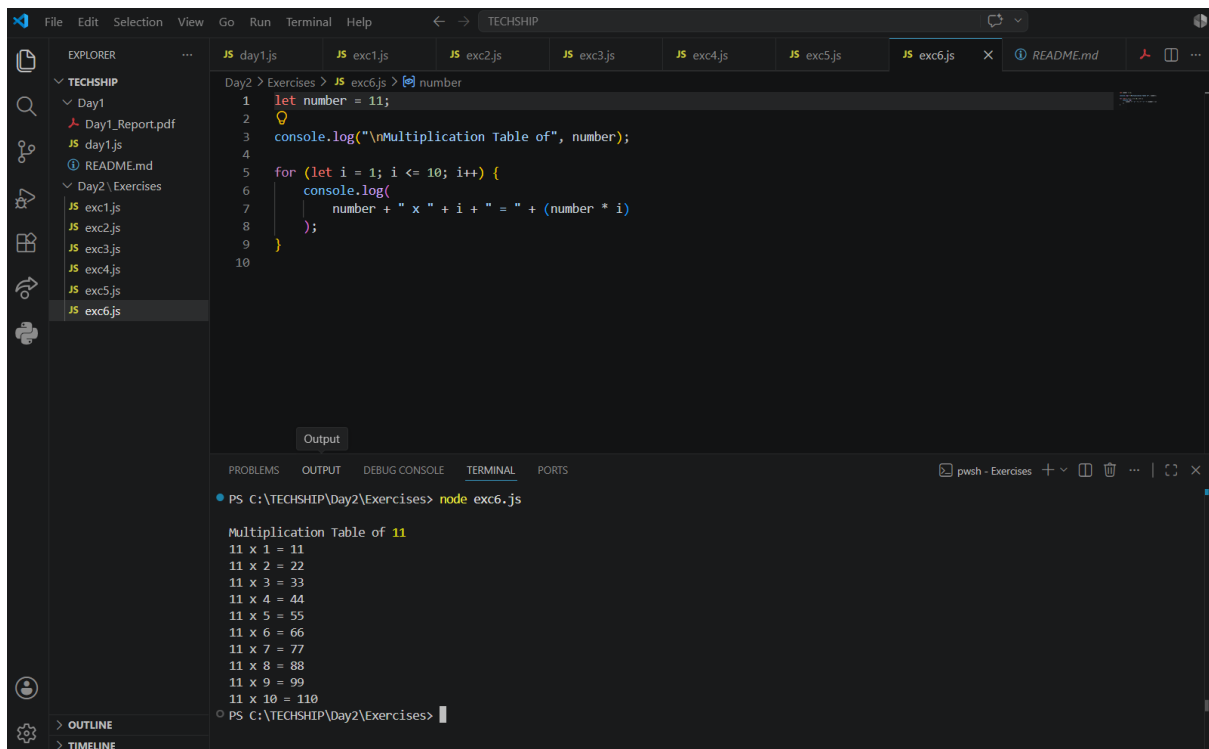
EXPLORER
TECHSHIP
  Day1
    Day1_Report.pdf
    day1.js
    README.md
  Day2 \ Exercises
    exc1.js
    exc2.js
    exc3.js
    exc4.js
    exc5.js

Day2 > Exercises > JS exc5.js > ...
1 let totalClasses = 200;
2 let attendedClasses = 175;
3
4 let attendancePercentage =
5   (attendedClasses / totalClasses) * 100;
6
7 console.log(
8   "\nAttendance Percentage:",
9   attendancePercentage + "%"
10 );

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\TECHSHIP\Day2\Exercises> node exc5.js

Attendance Percentage: 87.5%
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 6 – Multiplication Table Generator



```
File Edit Selection View Go Run Terminal Help
TECHSHIP

EXPLORER
TECHSHIP
  Day1
    Day1_Report.pdf
    day1.js
    README.md
  Day2 \ Exercises
    exc1.js
    exc2.js
    exc3.js
    exc4.js
    exc5.js
    exc6.js

Day2 > Exercises > JS exc6.js > number
1 let number = 11;
2
3 console.log("\nMultiplication Table of", number);
4
5 for (let i = 1; i <= 10; i++) {
6   console.log(
7     number + " x " + i + " = " + (number * i)
8   );
9 }
10

Output

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\TECHSHIP\Day2\Exercises> node exc6.js

Multiplication Table of 11
11 x 1 = 11
11 x 2 = 22
11 x 3 = 33
11 x 4 = 44
11 x 5 = 55
11 x 6 = 66
11 x 7 = 77
11 x 8 = 88
11 x 9 = 99
11 x 10 = 110
PS C:\TECHSHIP\Day2\Exercises>
```

Exercise 7 – Sum of Natural Numbers

The screenshot shows a Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a file structure with 'Day2' containing an 'Exercises' folder. The 'exc7.js' file is selected and open in the editor. The code in 'exc7.js' is as follows:

```
Day2 > Exercises > JS exc7.js > ...
1  let sum = 0;
2
3  for (let i = 1; i <= 100; i++) {
4      sum += i;
5  }
6
7  console.log("\nSum of numbers from 1 to 100 =", sum);
8
```

The TERMINAL panel at the bottom shows the command prompt output:

```
PS C:\TECHSHIP\Day2\Exercises> node exc7.js

Sum of numbers from 1 to 100 = 5050
PS C:\TECHSHIP\Day2\Exercises>
```

Challenge Problems

Challenge 1 – Even and Odd Number Generator

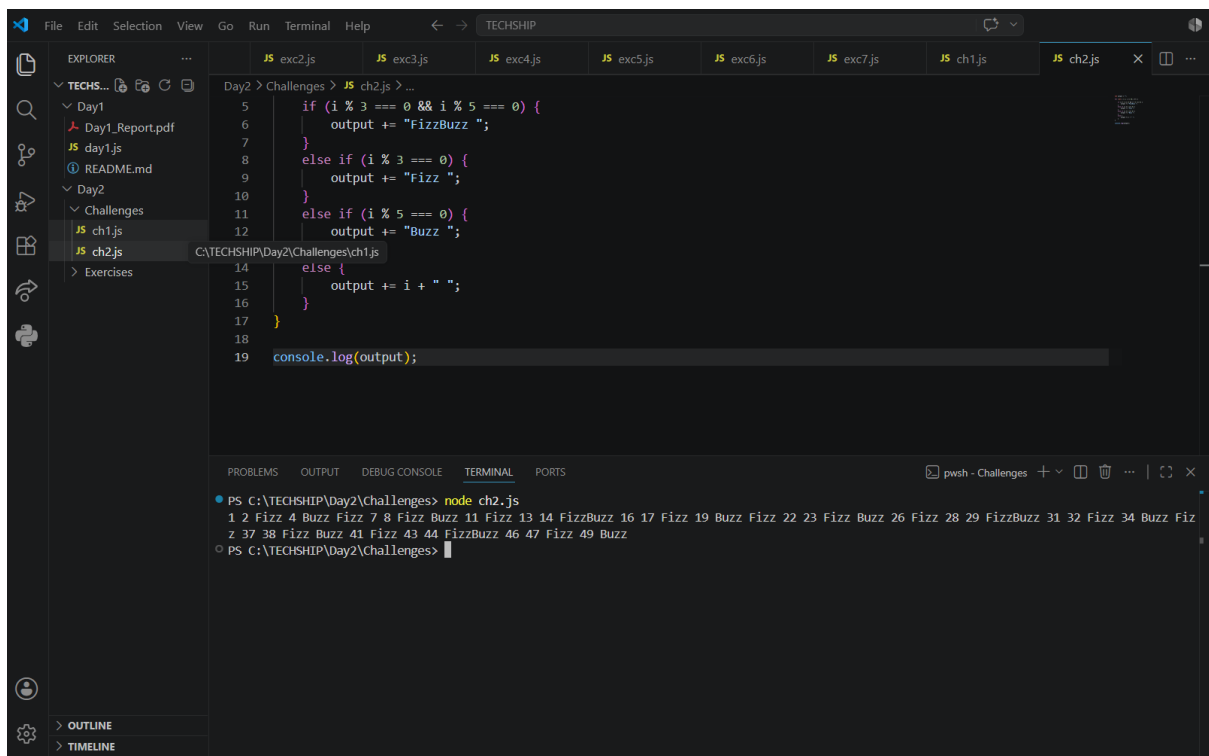
The screenshot shows a Visual Studio Code editor with a project named 'TECHSHIP'. The Explorer sidebar on the left shows a file structure with 'Day2' containing a 'Challenges' folder. The 'ch1.js' file is selected and open in the editor. The code in 'ch1.js' is as follows:

```
Day2 > Challenges > JS ch1.js > ...
1  let evenNumbers = "";
2  let oddNumbers = "";
3
4  for (let i = 1; i <= 100; i++) {
5      if (i % 2 === 0) {
6          evenNumbers += i + " ";
7      } else {
8          oddNumbers += i + " ";
9      }
10 }
11
12 console.log("Even Numbers:", evenNumbers);
13 console.log("Odd Numbers :", oddNumbers);
```

The TERMINAL panel at the bottom shows the command prompt output:

```
PS C:\TECHSHIP> cd Day2/challenges
PS C:\TECHSHIP\Day2\challenges> node ch1.js
Even Numbers: 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100
Odd Numbers : 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99
PS C:\TECHSHIP\Day2\challenges>
```

Challenge 2 - FizzBuzz



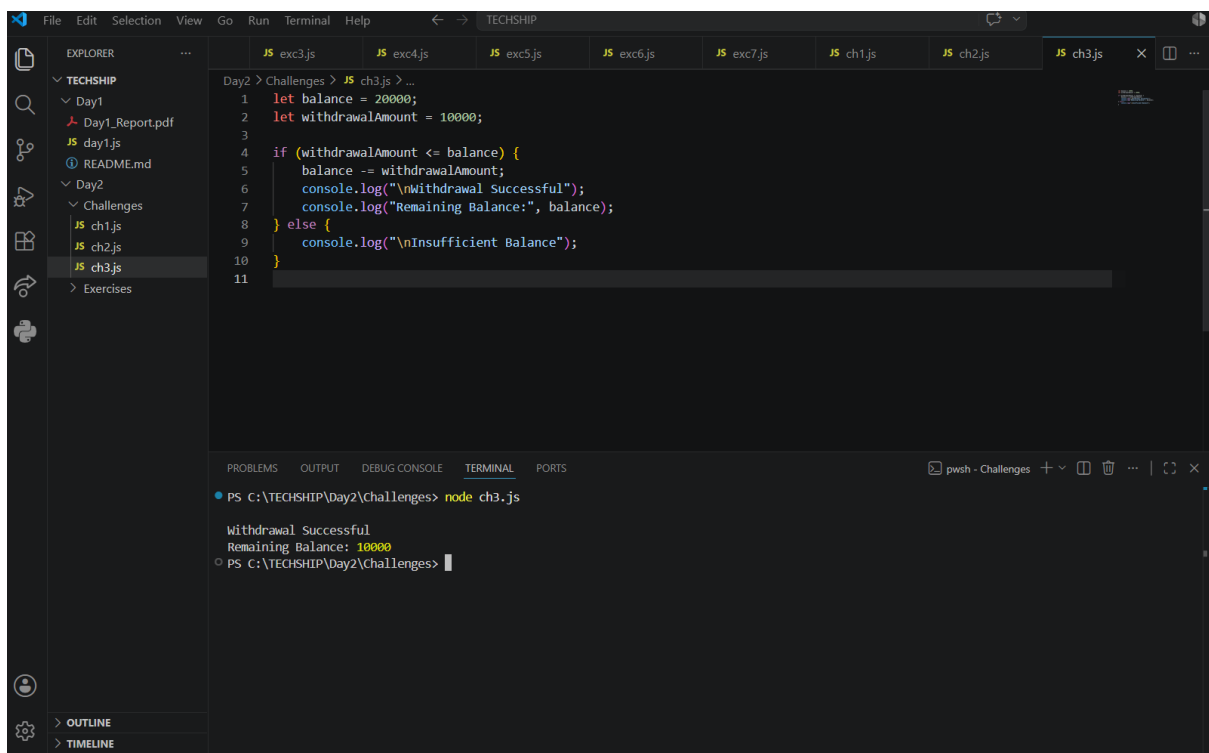
The screenshot shows the Visual Studio Code editor with the file explorer on the left. The file explorer shows a project named 'TECHSHIP' with a folder 'Day2' containing a subfolder 'Challenges'. Inside 'Challenges', there are files 'ch1.js' and 'ch2.js'. The file 'ch2.js' is selected and its content is displayed in the editor. The code is a JavaScript function that takes an array of numbers and returns an array of strings based on the FizzBuzz rules. The terminal at the bottom shows the command 'node ch2.js' being executed, and the output is a single line of 34 numbers and strings separated by spaces, representing the FizzBuzz sequence from 1 to 34.

```
Day2 > Challenges > JS ch2.js > ...
5     if (i % 3 === 0 && i % 5 === 0) {
6         output += "FizzBuzz ";
7     }
8     else if (i % 3 === 0) {
9         output += "Fizz ";
10    }
11    else if (i % 5 === 0) {
12        output += "Buzz ";
13    }
14    else {
15        output += i + " ";
16    }
17 }
18
19 console.log(output);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\TECHSHIP\Day2\Challenges> node ch2.js
1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz 16 17 Fizz 19 Buzz Fizz 22 23 Fizz Buzz 26 Fizz 28 29 FizzBuzz 31 32 Fizz 34 Buzz Fizz 37 38 Fizz Buzz 41 Fizz 43 44 FizzBuzz 46 47 Fizz 49 Buzz

Challenge 3 – ATM Withdrawal System



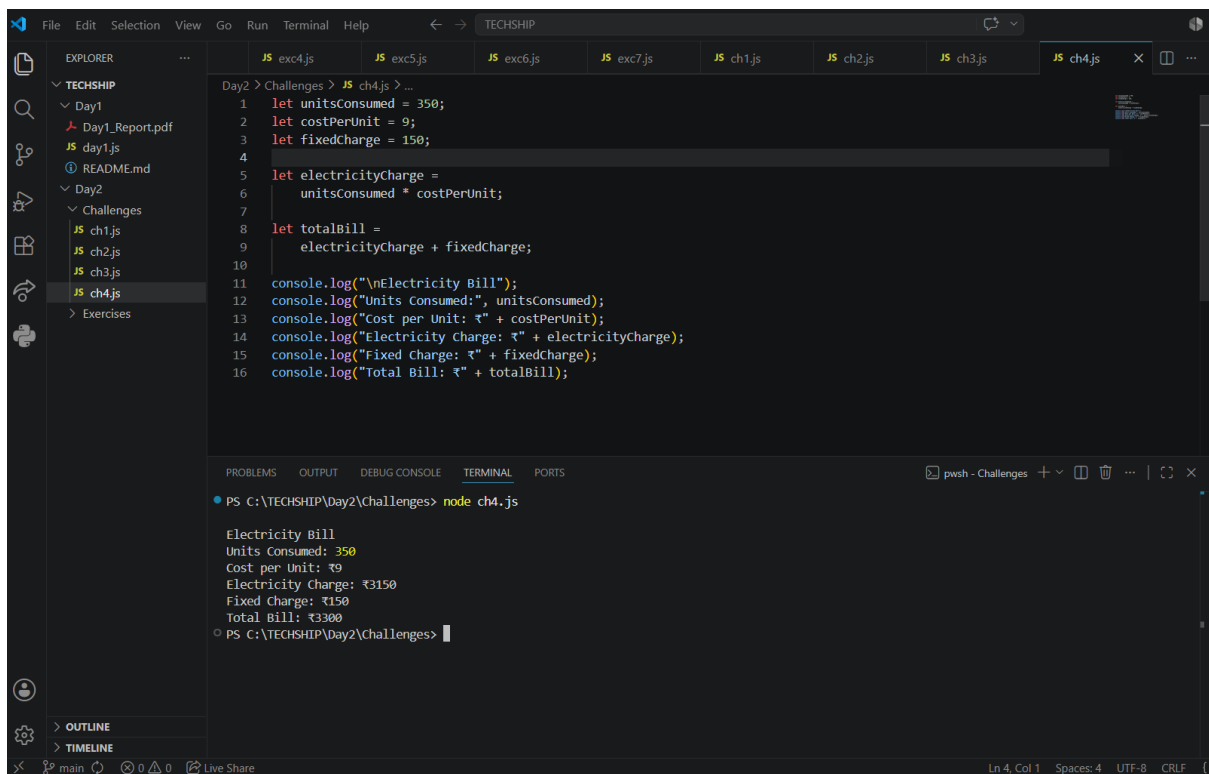
The screenshot shows the Visual Studio Code editor with the file explorer on the left. The file explorer shows a project named 'TECHSHIP' with a folder 'Day2' containing a subfolder 'Challenges'. Inside 'Challenges', there are files 'ch1.js', 'ch2.js', and 'ch3.js'. The file 'ch3.js' is selected and its content is displayed in the editor. The code is a JavaScript function that takes a balance and a withdrawal amount and returns a string indicating the result of the withdrawal. The terminal at the bottom shows the command 'node ch3.js' being executed, and the output is 'Withdrawal Successful' followed by 'Remaining Balance: 10000'.

```
Day2 > Challenges > JS ch3.js > ...
1     let balance = 20000;
2     let withdrawalAmount = 10000;
3
4     if (withdrawalAmount <= balance) {
5         balance -= withdrawalAmount;
6         console.log("\nWithdrawal Successful");
7         console.log("Remaining Balance:", balance);
8     } else {
9         console.log("\nInsufficient Balance");
10    }
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\TECHSHIP\Day2\Challenges> node ch3.js
Withdrawal Successful
Remaining Balance: 10000

Challenge 4 – Electricity Bill Calculator



```
File Edit Selection View Go Run Terminal Help < -> TECHSHIP

EXPLORER
TECHSHIP
  Day1
    Day1_Report.pdf
    JS day1.js
    README.md
  Day2
    Challenges
      JS ch1.js
      JS ch2.js
      JS ch3.js
      JS ch4.js
    Exercises

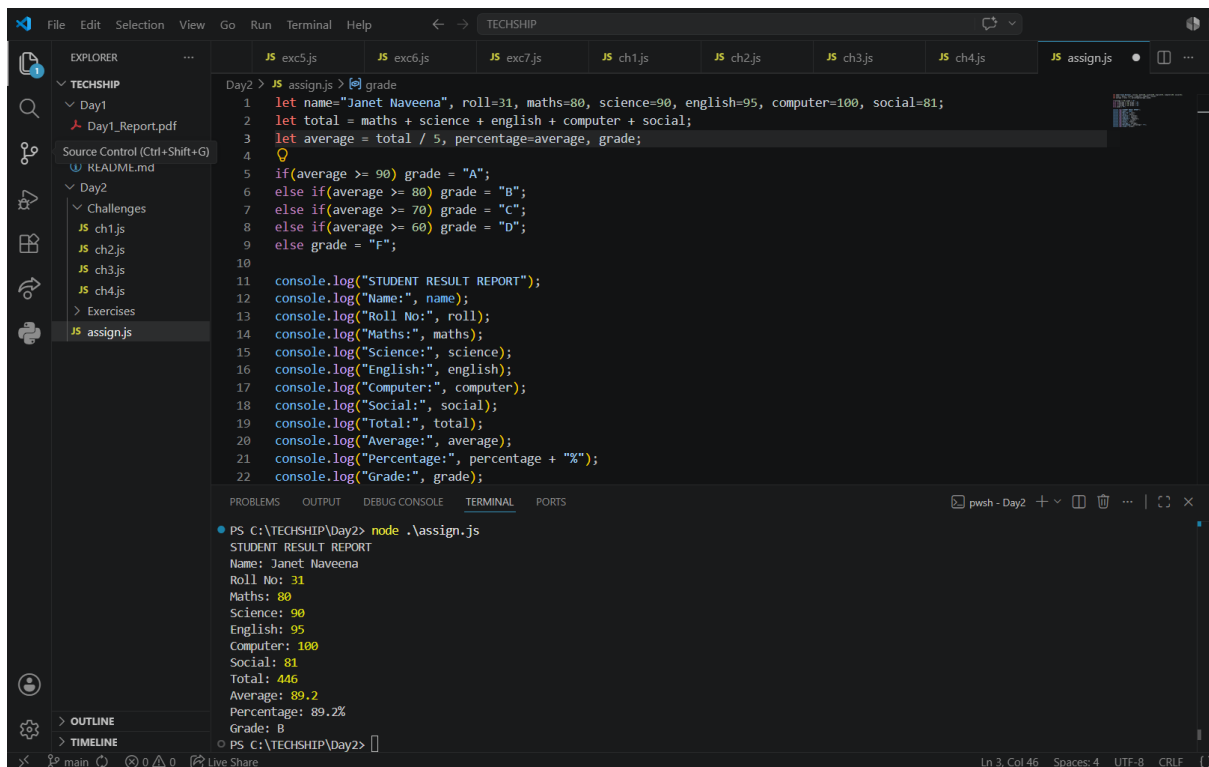
Day2 > Challenges > JS ch4.js > ...
1 let unitsConsumed = 350;
2 let costPerUnit = 9;
3 let fixedCharge = 150;
4
5 let electricityCharge =
6   unitsConsumed * costPerUnit;
7
8 let totalBill =
9   electricityCharge + fixedCharge;
10
11 console.log("\nElectricity Bill");
12 console.log("Units Consumed:", unitsConsumed);
13 console.log("Cost per Unit: ₹" + costPerUnit);
14 console.log("Electricity Charge: ₹" + electricityCharge);
15 console.log("Fixed Charge: ₹" + fixedCharge);
16 console.log("Total Bill: ₹" + totalBill);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\TECHSHIP\Day2\Challenges> node ch4.js

Electricity Bill
Units Consumed: 350
Cost per Unit: ₹9
Electricity Charge: ₹3150
Fixed Charge: ₹150
Total Bill: ₹3300
PS C:\TECHSHIP\Day2\Challenges>
```

Assignment

Student Result Management System



```
File Edit Selection View Go Run Terminal Help < -> TECHSHIP

EXPLORER
TECHSHIP
  Day1
    Day1_Report.pdf
    README.md
  Day2
    Challenges
      JS ch1.js
      JS ch2.js
      JS ch3.js
      JS ch4.js
    Exercises
      JS assign.js

Day2 > JS assign.js > grade
1 let name="Janet Naveena", roll=31, maths=80, science=90, english=95, computer=100, social=81;
2 let total = maths + science + english + computer + social;
3 let average = total / 5, percentage=average, grade;
4
5 if(average >= 90) grade = "A";
6 else if(average >= 80) grade = "B";
7 else if(average >= 70) grade = "C";
8 else if(average >= 60) grade = "D";
9 else grade = "F";
10
11 console.log("STUDENT RESULT REPORT");
12 console.log("Name:", name);
13 console.log("Roll No:", roll);
14 console.log("Maths:", maths);
15 console.log("Science:", science);
16 console.log("English:", english);
17 console.log("Computer:", computer);
18 console.log("Social:", social);
19 console.log("Total:", total);
20 console.log("Average:", average);
21 console.log("Percentage:", percentage + "%");
22 console.log("Grade:", grade);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\TECHSHIP\Day2> node .\assign.js

STUDENT RESULT REPORT
Name: Janet Naveena
Roll No: 31
Maths: 80
Science: 90
English: 95
Computer: 100
Social: 81
Total: 446
Average: 89.2
Percentage: 89.2%
Grade: B
PS C:\TECHSHIP\Day2>
```